Energy yield loss during communication outage

Wind Farm T (demo) · 01.09.2026 – 03.09.2026 · turbines T1, T2, T3

ESTIMATED LOST ENERGY (P50)

311.4 MWh

LIKELY RANGE (P10 – P90)

270.0 – 359.9 MWh

Over 72 h for 3 turbines. Value at 73.50 EUR/MWh: EUR 22,887 (EUR 19,846 – 26,455).

1 Outage details

Wind farm	Wind Farm T (demo)
Operator / insured	Demo Operator GmbH
Insurer · policy	Demo Insurance AG · DEMO-0000-00
Outage start	01 September 2026 at 00:00 (local time, Europe/Berlin)
Outage end	04 September 2026 at 00:00 (local time, Europe/Berlin)
Duration	72 hours
Affected turbines	T1, T2, T3 (4.5 MW class)
Suspected cause	A production stop is suspected; possible causes include cable damage between the wind farm and the grid connection point or a technical fault preventing feed-in. The root cause is under investigation.
Waiting period	24 h → 188.1 MWh (P50) beyond the waiting period, EUR 13,829

2 Lost energy per turbine

Turbine	P10 MWh	P50 MWh	P90 MWh	Avg wind m/s	Avg power MW
T1	74.6	89.1	104.8	5.9	1.24
T2	111.7	123.6	144.7	6.7	1.72
T3	83.7	98.6	110.4	6.2	1.37
Wind farm	270.0	311.4	359.9		

P50 is the best estimate; actual lost energy is expected to fall between P10 and P90 in 8 out of 10 cases, based on the method's historical errors. Farm P10/P90 are summed per turbine (conservative, since all turbines share the same reference data). Power curve values are interpreted as kW; energy = expected power × 10 min.

3 Reconstructed wind and lost production

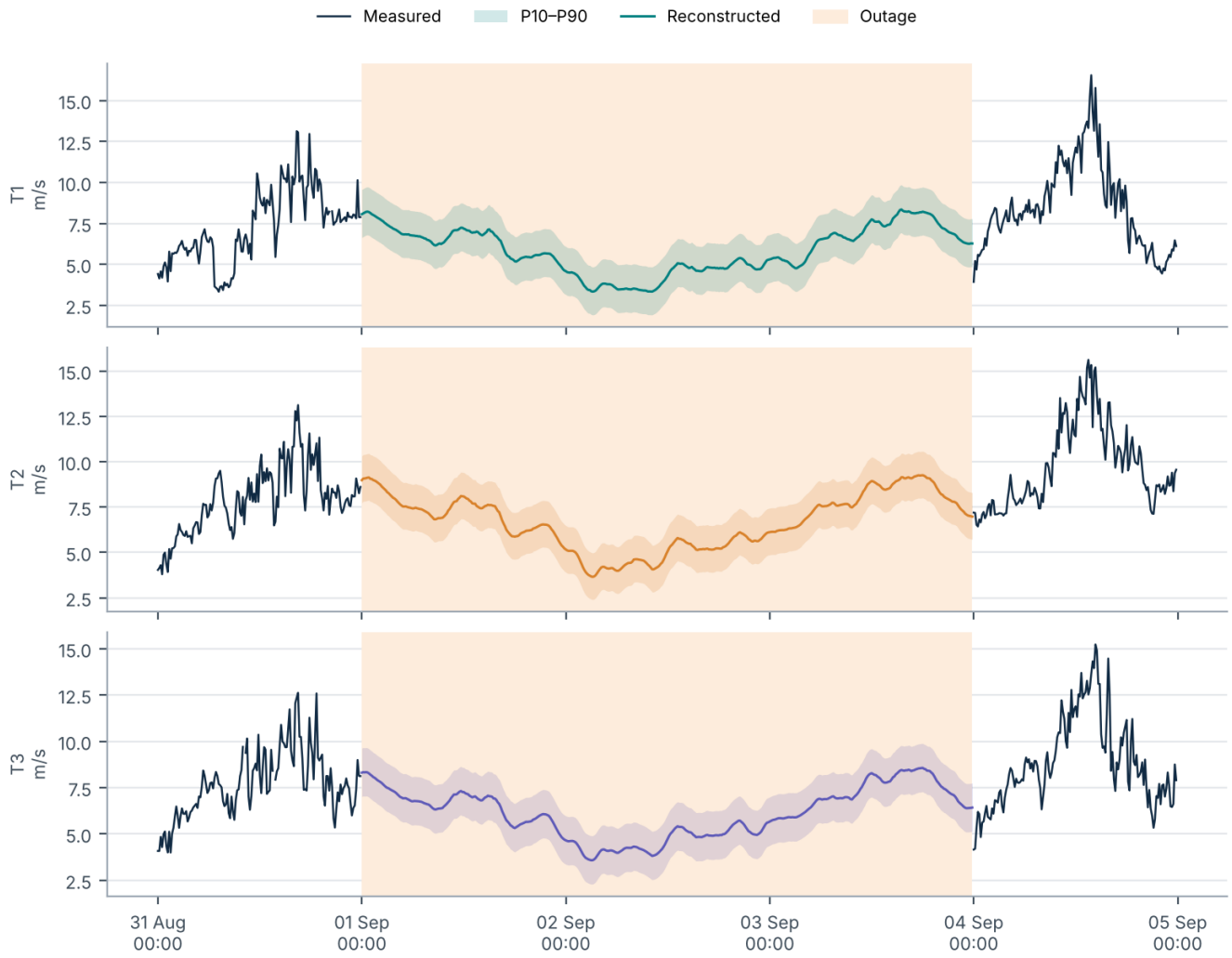


Figure 1 – Measured wind speed at each turbine (navy) one day before and after the outage, and the reconstructed wind speed during the outage (shaded) with its P10–P90 band.

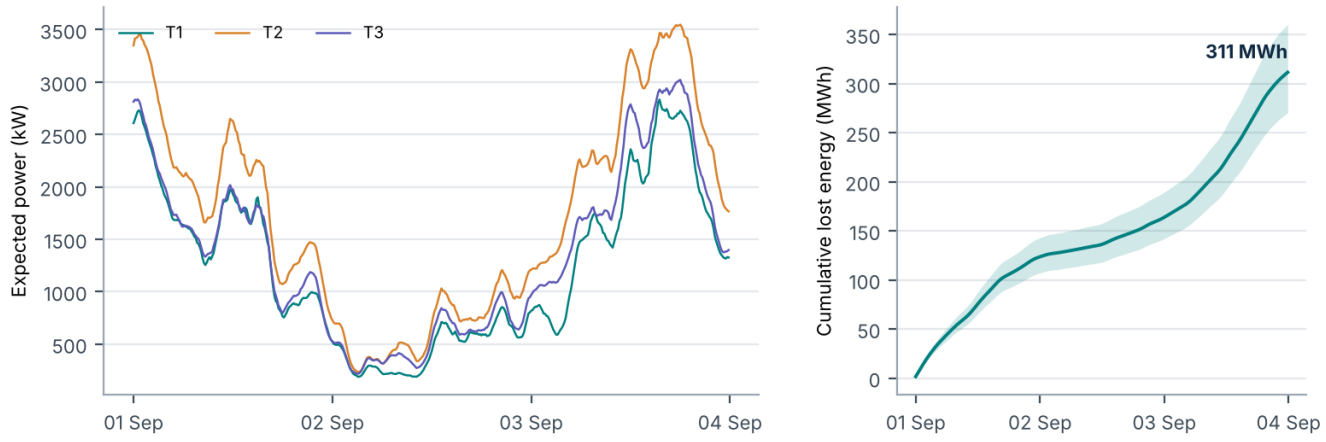


Figure 2 – Left: expected power per turbine during the outage. Right: cumulative lost energy for the wind farm with P10–P90 band.

4 Method, accuracy and limitations

For each affected turbine, the wind speed during the outage was reconstructed as a weighted sum of wind speeds measured at reference turbines in other wind farms (H1, H2, H3, H4, H5, O1, O2). Each reference is first shifted by the time delay that best matches the target turbine and smoothed over 130 minutes; all weights are forced to be non-negative. References and extra terms (wind-direction dependent slope, 3-hour temperature change, turbulence) were kept only where they improved accuracy in cross-validation over 3-day blocks of history. Lost energy is the turbine power curve averaged over the model's historical error distribution.

Why this method: every number can be checked by hand. Each reference has one weight and one time shift (Appendix A); non-negative weights mean stronger wind at a reference can never lower the estimate. Inputs are selected automatically, but only kept if they measurably improve accuracy on held-out history, so the model stays as simple as the data allows.

Turbine	Wind RMSE	Wind MAE	Power RMSE	|Energy err.| %	Val. wind RMSE	Val. energy %
T1	1.23	0.92	705	12.9	1.50	+15.3
T2	1.07	0.81	628	8.4	1.17	-0.7
T3	1.09	0.82	622	8.6	1.29	-2.3

First four columns: cross-validation over 29 historical 3-day blocks (model refitted without each block, then scored on it). 'Val.' columns: the 4–9 Sep validation week, held out entirely. Wind in m/s, power in kW, energy error = deviation of summed energy.

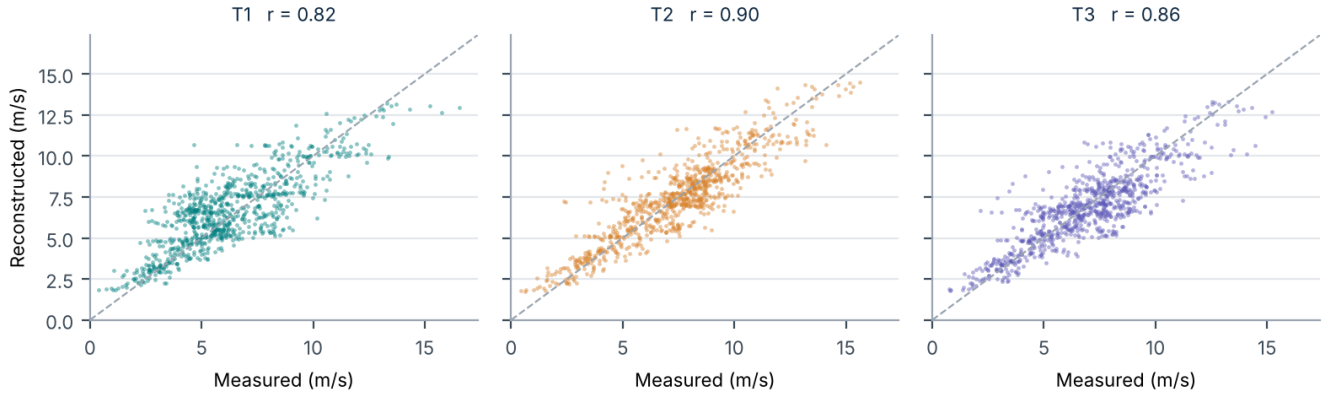


Figure 3 – Validation week: reconstructed vs measured 10-minute wind speed. Points on the dashed line would be perfect predictions.

Limitations

• The figure is an estimate. Individual 10-minute values are uncertain; totals over several days are considerably more reliable.
• It assumes the turbines were available and would have operated normally; curtailment, maintenance or grid limits during the outage are not included.
• Loss of communication does not by itself prove a production stop. The grid operator's metering data for the period should be used to confirm that no energy was fed in.
• Monetary value, waiting period and policy terms depend on contracts provided by the operator.

Appendix A Reproducibility and audit trail

Re-running the tool with the input files below (verified by their SHA-256 fingerprints) and the same model version reproduces every figure in this report.

Model	Bounded linear regression on lag-aligned reference turbines (git 49db391)
Reference turbines	H1, H2, H3, H4, H5, O1, O2
Outage window	2026-09-01 00:00 – 2026-09-03 23:50 (Europe/Berlin, 10-min stamps)
Power curve	rated 4,500 kW from 11.5 m/s; first output at 3 m/s
Report generated	24 Oct 2026 11:51

Turbine	H1	H2	H3	H4	H5	O1	O2
T1	0.36 (-20)	–	–	–	–	0.41 (+0)	–
T2	0.00 (-20)	–	0.14 (-30)	–	0.25 (-30)	0.17 (+0)	0.22 (+10)
T3	0.00 (-20)	0.14 (-30)	–	0.09 (-20)	0.18 (-30)	0.32 (+0)	–

Weight of each reference turbine's wind speed, with its time shift in minutes in brackets (negative = the wind reaches the target turbine first). Example: 0.40 means +1 m/s at that reference adds 0.40 m/s at the target turbine. '–' = reference not used for this turbine.

Turbine	Extra terms (coefficient)	Intercept m/s	CV R^2	Train rows
T1	ref_x_dir_sin +0.09, ref_x_dir_cos +0.03, turbulence +0.79	0.13	0.67	12212
T2	turbulence +0.68, temp_change_3h +0.03	0.11	0.77	12210
T3	turbulence +0.59	0.27	0.72	12256

Input file	SHA-256 fingerprint
H1_data.xlsx	`f00af7198126ac3a5839ab4cb1a09eb6ec54bf2bc98f7485b7ccd1afbf1334c5`
H2_data.xlsx	`5dfa885703128edcdddf2b4c7729f22219fd5803648f27c6a73737024bea0365`
H3_data.xlsx	`ddb005bfaf46af048cd5e752c63c78fae566c20f786be152872ba5178b3e2c4b`
H4_data.xlsx	`f4a213f36e8400ec36d444d2e53ef01d67658384f2078a4c8c334c861364caf6`
H5_data.xlsx	`ffa19bb61b99c3ae5347dac1c671e20955c70f466d1851466edf7b58cfc1722e`
O1_data.xlsx	`7139e3380bb1f1acc4f9094a7e372f03f50d7fbd06178c2a4ac269a7e541bfd7`
O2_data.xlsx	`d12d19e7a0c496c213cb24ce0a88c9290b212404b70b21b437a021857d0ea292`
T1_data.xlsx	`56e4139af0334f3e28393ac5757b545323790fc89043b0d4cbda08d400731b55`
T1_power_curve.xlsx	`cd7fbbf142148d9fcd84e95cfb9e2404c2e9a9bd399d13cf3dc669a4e123a6c1`
T2_data.xlsx	`3a6bb6e248a8ce67fd224a85ac433b09b48d89479b6044a92f43569f9a76b9e3`
T2_power_curve.xlsx	`2cfcffb092bb7412deb12ee0332b4e7d8200ea9768f211cd0961c18ee03f389e`
T3_data.xlsx	`2aba8062d668474ace743c0fc4c79c802137af72d63f43edd0d91b3e30d5fc4a`
T3_power_curve.xlsx	`7434f64acf9b77bfefd084f4b26d25a0cd86a136206bf21048ef4548a50a6f02`